

VIT CHENNAI · BMAT102L

Differential Equations & Transforms

Complete Study Guide · 7 Modules · Formula Sheets · Solved Problems

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Second Order Linear ODE — CF & PI

The general solution of a 2nd-order linear ODE is $GS = CF + PI$. The Complementary Function (CF) solves the homogeneous equation via the Auxiliary Equation (AE). The Particular Integral (PI) handles the forcing function using the operator method.

1.1 Complementary Function — Three Cases

CF BASED ON AE ROOTS ($m^2 + am + b = 0$)

REAL & DISTINCT ROOTS $m_1 \neq m_2$

$$CF = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

REAL & REPEATED $m_1 = m_2 = m$

$$CF = (c_1 + c_2 x) e^{mx}$$

COMPLEX ROOTS $\alpha \pm i\beta$

$$CF = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

1 CF — all three cases illustrated

Solve: (a) $y'' - 5y' + 6y = 0$ (b) $y'' - 6y' + 9y = 0$ (c) $y'' - 4y' + 13y = 0$

SOLUTION

a AE: $m^2 - 5m + 6 = (m - 2)(m - 3) = 0 \Rightarrow m = 2, 3$. CF: $c_1 e^{2x} + c_2 e^{3x}$

b AE: $(m - 3)^2 = 0 \Rightarrow m = 3$ (repeated). CF: $(c_1 + c_2 x) e^{3x}$

c AE: $m = \frac{4 \pm \sqrt{16 - 52}}{2} = 2 \pm 3i$. CF: $e^{2x} (c_1 \cos 3x + c_2 \sin 3x)$

1.2 Particular Integral — Operator Method

PI FORMULAS

PI FOR E^{ax} ($F(A) \neq 0$)

PI FOR E^{ax} (FAILURE: $F(A) = 0$)

PI FOR $\sin(Ax)/\cos(Ax)$

Substitute $D^2 = -a^2$; rationalise

$$\text{PI} = \frac{e^{ax}}{f(a)}$$

Multiply by x ; use $\frac{xe^{ax}}{f'(a)}$

PI FOR X^n

Expand $[f(D)]^{-1}$ as binomial series; operate on x^n

SHIFT RULE

$$\frac{1}{f(D)} e^{ax} V = e^{ax} \frac{1}{f(D+a)} V$$

GS

GS = CF + PI (always combine both)

1 Complete solution (CF + PI)

Solve $(D^2 - 3D + 2)y = e^{4x}$.

SOLUTION

1 AE: $(m-1)(m-2) = 0 \Rightarrow m = 1, 2$. CF = $c_1 e^x + c_2 e^{2x}$

2 PI: $\frac{e^{4x}}{f(4)} = \frac{e^{4x}}{16-12+2} = \frac{e^{4x}}{6}$

$$y = c_1 e^x + c_2 e^{2x} + e^{4x}/6$$

2 PI for sinusoidal — rationalise

Find PI for $(D^2 + D + 1)y = \sin 2x$.

SOLUTION

1 Put $D^2 = -4$: PI = $\frac{\sin 2x}{-4+D+1} = \frac{\sin 2x}{D-3}$

2 Rationalise: $\frac{(D+3)}{D^2-9} \sin 2x = \frac{2 \cos 2x + 3 \sin 2x}{-4-9} = \frac{-(2 \cos 2x + 3 \sin 2x)}{13}$

$$\text{PI} = -(2\cos 2x + 3\sin 2x)/13$$

3 Failure case ($f(a) = 0$)

Find PI for $(D^2 - 5D + 6)y = e^{2x}$.

SOLUTION

1 $f(2)=4-10+6=0 \rightarrow$ failure. Use: $PI=\frac{xe^{2x}}{f'(2)}$. $f'(D) = 2D - 5$, $f'(2) = -1$

2 $PI=\frac{xe^{2x}}{-1} = -xe^{2x}$

$$PI = -xe^{2x}$$

Module 1 — Key Formulas

$$GS = CF + PI$$

Always combine both parts

$$PI: e^{ax} \text{ (f(a) \neq 0)}$$

$$e^{ax} / f(a)$$

$$\text{Failure PI}$$

$$xe^{ax} / f'(a)$$

Variation of Parameters & Euler-Cauchy

Variation of Parameters is a general method for PI when the operator method is insufficient. Euler-Cauchy equations have variable coefficients ($x^n y^{(n)}$ terms) and reduce to constant-coefficient equations via the substitution $x = e^t$.

2.1 Variation of Parameters

VARIATION OF PARAMETERS

For $y'' + Py' + Qy = R$ with CF $c_1 y_1 + c_2 y_2$:

$$PI = y_1 v_1 + y_2 v_2, \quad v_1 = - \int \frac{y_2 R}{W} dx, \quad v_2 = \int \frac{y_1 R}{W} dx$$

where $W = y_1 y_2' - y_2 y_1'$ is the **Wronskian**.

1 Variation of Parameters

Solve $y'' - 2y' + y = \frac{e^x}{x}$ using variation of parameters.

SOLUTION

- 1 CF: $(m - 1)^2 = 0 \Rightarrow y_1 = e^x, y_2 = x e^x$
- 2 Wronskian: $W = e^x(1 + x)e^x - x e^x \cdot e^x = e^{2x}$
- 3 $v_1 = - \int \frac{x e^x \cdot e^x / x}{e^{2x}} dx = - \int 1 dx = -x, v_2 = \int \frac{e^x \cdot e^x / x}{e^{2x}} dx = \int \frac{dx}{x} = \ln x$
- 4 $PI = e^x(-x) + x e^x \ln x = -x e^x + x e^x \ln x$

$$y = (c_1 + c_2 x) e^x + x e^x (\ln x - 1)$$

2 VoP with trig forcing

Solve $y'' + y = \sec x$.

SOLUTION

1 CF: $m = \pm i \Rightarrow y_1 = \cos x, y_2 = \sin x. W = \cos^2 x + \sin^2 x = 1$

2 $v_1 = -\int \sin x \sec x dx = -\int \tan x dx = \ln |\cos x|$

3 $v_2 = \int \cos x \sec x dx = x$

$y = c_1 \cos x + c_2 \sin x + \cos x \cdot \ln |\cos x| + x \sin x$

2.2 Euler-Cauchy Equations

EULER-Cauchy SUBSTITUTION

SUBSTITUTION

$x = e^t, t = \ln x$

$XY' = DY \ (D=D/DT)$

$xy' = Dy$

$X^2 Y'' = D(D-1)Y$

$x^2 y'' = D(D-1)y$

1 Euler-Cauchy homogeneous

Solve $x^2 y'' - 3xy' + 4y = 0$.

SOLUTION

1 Substitute: $[D(D-1) - 3D + 4]y = (D^2 - 4D + 4)y = (D-2)^2 y = 0$

2 AE: $(m-2)^2 = 0 \Rightarrow m = 2(\text{repeated}).$ In t : $y = (c_1 + c_2 t)e^{2t}$

3 Back-substitute $t = \ln x$: $y = (c_1 + c_2 \ln x)x^2$

$y = (c_1 + c_2 \ln x)x^2$

Module 2 — Key Formulas

Wronskian W

$$y_1 y_2' - y_2 y_1'$$

v_1 (VoP)

$$- \int y_2 R / W \, dx$$

Euler-Cauchy

$$x = e^t; \quad x^2 y'' = D(D-1)y$$

PDEs govern heat conduction, wave propagation, fluid flow, and quantum mechanics. VIT covers formation, Lagrange's first-order method, and separation of variables for the classical heat and wave equations.

3.1 Formation & Lagrange's Method

LAGRANGE'S METHOD FOR $Pp + Qq = R$

Auxiliary equations: $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$. Find two independent solutions $u = c_1$ and $v = c_2$. General solution: $\phi(u, v) = 0$.

1 Lagrange's method

Solve $xp + yq = z$.

SOLUTION

- 1 Auxiliary: $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$
- 2 From 1st and 2nd: $\ln x = \ln y + c \Rightarrow x/y = c_1$. From 2nd and 3rd: $y/z = c_2$

General solution: $\phi(x/y, y/z) = 0$ i.e., $z = y \cdot f(x/y)$

2 Form PDE by eliminating constants

Form PDE from $z = ax^2 + by^2$.

SOLUTION

- 1 $p = 2ax \Rightarrow a = p/(2x)$; $q = 2by \Rightarrow b = q/(2y)$
- 2 Substitute: $z = \frac{p}{2x}x^2 + \frac{q}{2y}y^2 = \frac{px+qy}{2}$

$$\text{PDE: } px + qy = 2z$$

3.2 Separation of Variables

STANDARD PDE FORMS

1D HEAT EQUATION

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$$

1D WAVE EQUATION

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

LAPLACE EQUATION

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

1 Heat equation — separation of variables

Solve $u_t = u_{xx}$ with $u(0, t) = u(\pi, t) = 0$, $u(x, 0) = \sin x$.

SOLUTION

- 1 Let $u = X(x)T(t)$: $XT' = X''T \Rightarrow \frac{T'}{T} = \frac{X''}{X} = -\lambda^2$
- 2 $X'' + \lambda^2 X = 0 \Rightarrow X = A \cos \lambda x + B \sin \lambda x$. BC: $X(0)=0 \rightarrow A=0$; $X(\pi)=0 \rightarrow \lambda=n$
- 3 $T' = -n^2 T \Rightarrow T = Ce^{-n^2 t}$. IC $u(x, 0) = \sin x \Rightarrow n = 1, C = 1$

$$u(x, t) = \sin(x) \cdot e^{-t}$$

Module 3 — Key Formulas

Lagrange aux.

$$dx/P = dy/Q = dz/R$$

Heat Eq.

$$u_t = \alpha^2 u_{xx}$$

Sep. variables

$$u = X(x) \cdot T(t)$$

Laplace Transforms

The Laplace transform converts ODE problems with initial conditions into algebraic equations in the s-domain, dramatically simplifying their solution. It's the standard tool in control systems, signal processing, and circuit analysis.

4.1 Standard Laplace Transform Table

LAPLACE TRANSFORM TABLE

$L\{1\}$ $1/s$	$L\{t^n\}$ $n!/s^{n+1}$	$L\{e^{at}\}$ $1/(s - a)$
$L\{\sin at\}$ $a/(s^2 + a^2)$	$L\{\cos at\}$ $s/(s^2 + a^2)$	$L\{e^{at} \sin bt\}$ $b/((s - a)^2 + b^2)$
$L\{e^{at} \cos bt\}$ $(s - a)/((s - a)^2 + b^2)$	$L\{f'(t)\}$ $sF(s) - f(0)$	$L\{f''(t)\}$ $s^2F(s) - sf(0) - f'(0)$

1 Solve IVP using Laplace Transform

Solve $y'' + 4y' + 3y = e^{-t}$ with $y(0) = 1, y'(0) = 0$.

STEP-BY-STEP SOLUTION

- Take LT: $(s^2Y - s + 4(sY - 1) + 3Y = \frac{1}{s+1})$
- $Y(s^2 + 4s + 3) = s + 4 + \frac{1}{s+1} = \frac{(s+4)(s+1)+1}{s+1} = \frac{s^2+5s+5}{s+1}$
- $Y = \frac{s^2+5s+5}{(s+1)^2(s+3)}$. Partial fractions: $\frac{A}{s+1} + \frac{B}{(s+1)^2} + \frac{C}{s+3}$
- Solving: $A=3/4, B=-1/2, C=1/4$. Inverse LT: $y = \frac{3}{4}e^{-t} - \frac{1}{2}te^{-t} + \frac{1}{4}e^{-3t}$

$$y = (3/4)e^{-t} - (1/2)te^{-t} + (1/4)e^{-3t}$$

2 Inverse Laplace by partial fractions

Find $\mathcal{L}^{-1}\left\{\frac{2s+3}{(s+1)(s+2)}\right\}$.

SOLUTION

1 Partial fractions: $\frac{A}{s+1} + \frac{B}{s+2}$. At $s=-1$: $A=1$. At $s=-2$: $B=1$.

2 Inverse: $e^{-t} + e^{-2t}$

$$e^{-t} + e^{-2t}$$

4.2 Shifting Theorems & Convolution

PROPERTIES

FIRST SHIFTING (S-SHIFT)

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$$

SECOND SHIFTING (T-SHIFT)

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$$

CONVOLUTION THEOREM

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

Module 4 — Key Formulas

$$\mathcal{L}\{e^{at}\}$$

$$1/(s-a)$$

$$\mathcal{L}\{f''\}$$

$$s^2F - sf(0) - f'(0)$$

$$\text{First shift}$$

$$\mathcal{L}\{e^{at}f\} = F(s-a)$$

Fourier Series, Transforms & Z-Transforms

Fourier series decompose periodic signals into sinusoids. Fourier transforms extend this to non-periodic signals. Z-transforms are the discrete-time counterpart, essential for digital signal processing and difference equations.

5.1 Fourier Series

FOURIER SERIES FORMULAS

GENERAL FORM $[-L, L]$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

EULER'S FORMULAE

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx; \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx; \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$$

EVEN FUNCTION $\rightarrow$ COSINE SERIES

$$\text{All } b_n = 0$$

ODD FUNCTION $\rightarrow$ SINE SERIES

$$\text{All } a_n = 0$$

PARSEVAL'S THEOREM

$$\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

FAMOUS RESULT

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad (\text{from } x^2 \text{ expansion})$$

1 Fourier series of $f(x) = x$

Find Fourier series of $f(x) = x$ in $(-\pi, \pi)$.

SOLUTION

1 $f(x)=x$ is odd $\rightarrow a_0=0, a_n=0. \quad b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) dx$

2 IBP: $b_n = \frac{2}{\pi} \left[-\frac{x \cos nx}{n} + \frac{\sin nx}{n^2} \right]_0^{\pi} = \frac{2(-1)^{n+1}}{n}$

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

5.2 Z-Transform

DEFINITION

$$Z\{x(n)\} = \sum_{n=0}^{\infty} x(n)z^{-n}$$

Z{U(N)} UNIT STEP

$$z/(z-1)$$

Z{Aⁿ}

$$z/(z-a)$$

Z{N·Aⁿ}

$$az/(z-a)^2$$

RIGHT SHIFT (DELAY)

$$Z\{x(n-k)\} = z^{-k}X(z)$$

INITIAL VALUE THEOREM

$$x(0) = \lim_{z \rightarrow \infty} X(z)$$

1 Find Z-transform

Find $Z\{3^n + 2n\}$.

SOLUTION

$$1 \quad Z\{3^n\} = \frac{z}{z-3}$$

$$2 \quad Z\{n\} = Z\{n \cdot 1^n\} = \frac{z}{(z-1)^2}$$

$$3 \quad X(z) = \frac{z}{z-3} + \frac{2z}{(z-1)^2}$$

$$X(z) = z/(z-3) + 2z/(z-1)^2$$

2 Inverse Z-transform

Find $Z^{-1}\left\{\frac{z^2}{(z-1)(z-2)}\right\}$.

SOLUTION

$$1 \quad \frac{X(z)}{z} = \frac{z}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{2}{z-2}$$

$$2 \quad X(z) = \frac{-z}{z-1} + \frac{2z}{z-2} \Rightarrow x(n) = -1 + 2 \cdot 2^n = 2^{n+1} - 1$$

$$x(n) = 2^{n+1} - 1$$

Module 5 — Key Formulas

Fourier b_n

$$\frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx$$

Z{aⁿ}

$$z/(z - a)$$

Z shift

$$Z\{x(n - k)\} = z^{-k}X(z)$$

Applications of Differential Equations

Differential equations model virtually every physical system. Spring-mass systems, electrical circuits, population growth, and heat flow all lead to ODEs or PDEs that can be solved using the techniques from previous modules.

6.1 Spring-Mass System

MECHANICAL VIBRATIONS

NEWTON'S 2ND LAW

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

UNDAMPED FREE (C=0, F=0)

$$\ddot{x} + \omega_n^2 x = 0; \omega_n = \sqrt{k/m}$$

NATURAL FREQUENCY

$$f_n = \omega_n / (2\pi) \text{ Hz}$$

DAMPING RATIO

$$\zeta = c / (2\sqrt{mk})$$

OVER/UNDER/CRITICAL DAMPED

$$\zeta > 1 / \zeta < 1 / \zeta = 1$$

RESONANCE

Forced frequency = ω_n ; amplitude grows unbounded (undamped)

1 Undamped spring-mass

A 2 kg mass on a spring ($k = 50 \text{ N/m}$) is displaced 0.1 m and released. Find $x(t)$.

SOLUTION

- 1 ODE: $2\ddot{x} + 50x = 0 \Rightarrow \ddot{x} + 25x = 0$. $\omega_n = 5 \text{ rad/s}$
- 2 General solution: $x = A \cos 5t + B \sin 5t$. IC: $x(0) = 0.1 \Rightarrow A = 0.1$; $\dot{x}(0) = 0 \Rightarrow B = 0$

$$x(t) = 0.1 \cos(5t) \text{ m}$$

6.2 RLC Circuit

RLC CIRCUIT ODE

KIRCHHOFF'S VOLTAGE LAW

$$L \frac{di}{dt} + Ri + \frac{q}{C} = E(t); \text{ differentiating: } L\ddot{q} + R\dot{q} + \frac{q}{C} = E(t)$$

ANALOGIES

$$L \leftrightarrow m, R \leftrightarrow c, 1/C \leftrightarrow k, E(t) \leftrightarrow F(t), q \leftrightarrow x$$

1 RLC circuit solution

An RLC series circuit has $R=4\Omega$, $L=1H$, $C=0.2F$, $E=0$. Find $q(t)$ if $q(0)=1$, $q'(0)=0$.

SOLUTION

1 ODE: $\ddot{q} + 4\dot{q} + 5q = 0$. AE: $m^2 + 4m + 5 = 0 \Rightarrow m = -2 \pm i$

2 GS: $q = e^{-2t}(c_1 \cos t + c_2 \sin t)$. IC $q(0)=1$: $c_1=1$. $q'(0)=0$: $-2c_1 + c_2 = 0 \rightarrow c_2=2$

$$q(t) = e^{-2t}(\cos t + 2 \sin t)$$

6.3 Population Models & Newton's Law of Cooling

FIRST-ORDER APPLICATION MODELS

EXPONENTIAL GROWTH

$$\frac{dP}{dt} = kP \Rightarrow P = P_0 e^{kt}$$

LOGISTIC GROWTH

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M}\right)$$

NEWTON'S LAW OF COOLING

$$\frac{dT}{dt} = k(T - T_\infty) \Rightarrow T = T_\infty + (T_0 - T_\infty)e^{kt}$$

1 Newton's law of cooling

A cup of coffee at 90°C cools in a 20°C room. After 5 min, temperature is 70°C . Find k and temperature after 10 min.

SOLUTION

1 $T = 20 + (90 - 20)e^{kt} = 20 + 70e^{kt}$. At $t=5$: $70 = 20 + 70e^{5k} \Rightarrow e^{5k} = 5/7 \Rightarrow k = \frac{\ln(5/7)}{5} \approx -0.0673$

2 At $t=10$: $T = 20 + 70e^{10k} = 20 + 70(5/7)^2 = 20 + 70(25/49) = 20 + 35.7 = 55.7^\circ C$

$k \approx -0.0673 \text{ min}^{-1}$; $T(10) \approx 55.7^\circ C$

Module 6 — Key Models

Spring-mass

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

RLC analogies

$$L \leftrightarrow m, R \leftrightarrow c, 1/C \leftrightarrow k$$

Newton's Cooling

$$T = T_\infty + (T_0 - T_\infty)e^{kt}$$

When ODEs lack closed-form solutions, numerical methods compute accurate approximate solutions step by step. Euler's method is simple; Runge-Kutta 4th order (RK4) is the workhorse of scientific computing, balancing accuracy and efficiency.

7.1 Euler's Method

EULER'S METHOD

ITERATION FORMULA

$y_{n+1} = y_n + h \cdot f(x_n, y_n)$ where h is step size

ERROR (LOCAL)

$O(h^2)$ per step; $O(h)$ globally — first-order method

1 Euler's method

Solve $y' = x + y$ with $y(0) = 1$. Use $h=0.1$ for 3 steps.

SOLUTION

- 1 $f(x,y)=x+y$. Step 1: $y_1 = 1 + 0.1(0 + 1) = 1.1$
- 2 Step 2: $y_2 = 1.1 + 0.1(0.1 + 1.1) = 1.1 + 0.12 = 1.22$
- 3 Step 3: $y_3 = 1.22 + 0.1(0.2 + 1.22) = 1.22 + 0.142 = 1.362$
- 4 Exact at $x=0.3$: $y = 2e^{0.3} - 0.3 - 1 = 2(1.3499) - 1.3 = 1.3998 \approx 1.40$. Error $\approx 3\%$

$y(0.1) \approx 1.1$, $y(0.2) \approx 1.22$, $y(0.3) \approx 1.362$

7.2 Runge-Kutta 4th Order (RK4)

FOUR SLOPES

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_n + h/2, y_n + k_1/2)$$

$$k_3 = hf(x_n + h/2, y_n + k_2/2)$$

$$k_4 = hf(x_n + h, y_n + k_3)$$

UPDATE

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Error: $O(h^4)$ per step; $O(h^5)$ globally — very accurate

1 RK4 — one step

Apply RK4 to $y' = x + y$, $y(0) = 1$ with $h=0.1$ to find $y(0.1)$.

STEP-BY-STEP SOLUTION

$$1 \quad k_1 = 0.1 \cdot f(0, 1) = 0.1(0 + 1) = 0.1$$

$$2 \quad k_2 = 0.1 \cdot f(0.05, 1.05) = 0.1(0.05 + 1.05) = 0.11$$

$$3 \quad k_3 = 0.1 \cdot f(0.05, 1.055) = 0.1(0.05 + 1.055) = 0.1105$$

$$4 \quad k_4 = 0.1 \cdot f(0.1, 1.1105) = 0.1(0.1 + 1.1105) = 0.12105$$

$$5 \quad y_1 = 1 + \frac{1}{6}(0.1 + 0.22 + 0.221 + 0.12105) = 1 + \frac{0.66205}{6} = 1 + 0.11034 = 1.11034$$

$$6 \quad \text{Exact: } 2e^{0.1} - 1.1 = 1.11034. \text{ RK4 is exact to 5 decimal places!}$$

$y(0.1) \approx 1.11034$ (exact to 5 decimal places)

Module 7 — Key Formulas

Euler

$$y_{n+1} = y_n + hf(x_n, y_n)$$

RK4 update

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

RK4 accuracy

$O(h^4)$ per step — very accurate

BMAT102L — Master Formula Sheet

Modules 1–2: Linear ODE

CF DISTINCT

$$c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

CF COMPLEX $A \pm iB$

$$e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

PI: E^{ax}

$$e^{ax} / f(a)$$

WRONSKIAN

$$W = y_1 y_2' - y_2 y_1'$$

EULER-CAUCHY

$$x = e^t; \quad x^2 y'' = D(D-1)y$$

SHIFT RULE

$$e^{ax} f(D+a)^{-1} V$$

Modules 3–4: PDE & Laplace

LAGRANGE AUX

$$dx/P = dy/Q = dz/R$$

HEAT EQ.

$$u_t = \alpha^2 u_{xx}$$

L{SIN AT}

$$a / (s^2 + a^2)$$

L{F''}

$$s^2 F - s f(0) - f'(0)$$

1ST SHIFT

$$\mathcal{L}\{e^{at} f\} = F(s-a)$$

CONVOLUTION

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

Modules 5–7: Transforms & Numerics

FOURIER B_n

$$\frac{1}{L} \int f \sin \frac{n\pi x}{L} dx$$

Z{Aⁿ}

$$z / (z - a)$$

NEWTON'S COOLING

$$T = T_\infty + (T_0 - T_\infty) e^{kt}$$

EULER METHOD

$$y_{n+1} = y_n + h f(x_n, y_n)$$

RK4

$$\frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

SPRING-MASS

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

BMAT102L — Complete

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